BASIC ANTIVIRUS SIMULATION (SIGNATURE SCANNER)

Project Report

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1. Introduction

Cybersecurity has become an essential requirement in today's digital world. Malware such as viruses, worms, trojans, and ransomware can compromise systems and steal sensitive information. Antivirus software is designed to identify and remove such threats before they can cause damage. This project demonstrates the basic working principle of a signature-based antivirus system.

2. Problem Statement

The system scans files, generates SHA-256 hashes, compares them with known malware signatures, detects malicious files, quarantines infected files, and generates logs and reports.

3. Objectives

• Understand signature-based malware detection
• Learn SHA-256 hashing
• Identify malicious files
• Implement quarantine functionality
• Generate logs and reports

4. Tools and Technologies Used

Python, VS Code, hashlib, os, shutil, datetime, Windows OS

5. Project Architecture

Files → Hash Generation → Signature Comparison → Malware Detection → Quarantine → Logging → Report Generation

6. Code Explanation

The project uses SHA-256 hashing to create unique fingerprints of files. Hashes are compared against known malware signatures. Matching files are flagged, quarantined, logged, and included in a final report.

7. Working Process

1. Load malware signatures.
2. Scan files.
3. Generate SHA-256 hashes.
4. Compare hashes with signatures.
5. Detect malware.
6. Quarantine infected files.
7. Create logs.
8. Generate final report.

8. Results

The antivirus simulation successfully scanned files, detected malicious files using signatures, quarantined infected files, generated logs, and created reports.

9. Conclusion

This project provided practical experience in malware detection, hashing, logging, and incident response concepts using Python.

10. Future Enhancements

Real-time monitoring, GUI interface, cloud signature updates, email alerts, scheduled scanning, heuristic detection, and threat intelligence integration.

11. References

Python Documentation, NIST SHA-256 Standards, Cybersecurity Fundamentals, Malware Detection Concepts.